

from supercomputers to cell phones. Graphene-based nanoelectronic devices will not only be faster and less power-consuming than silicon, these will also be much thinner.

Graphene, a form of carbon, consists of a layer of graphite 50,000 times thinner than a human hair with unique electronic properties.

Victor Aristov of the Leibniz Institute for Solid State and Materials Research, Germany, and colleagues have grown high-quality graphene on the surface of commercially available silicon carbide wafers to produce material with excellent electronic properties. It is claimed to be a huge step toward technological application of this material as the synthesis is compatible with industrial mass production.

## Transistor integrated with cell membrane

In an experiment that is being touted as the most intimate linking between man and machine till date, scientists have implanted a nano-sized transistor inside a cell-like membrane and powered it using the cell's own fuel. The experiment by University of California scientists has paved the way for new types of man-machine interactions, and may help experts learn the inner workings of disease-related proteins inside cell membranes, leading to new ways to read and control brain or nerve cells.

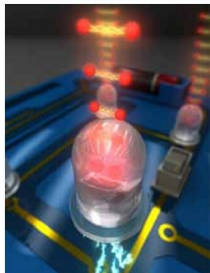
For the transistor, the scientists used a next-generation material known as a carbon nanotube, instead of most commonly used silicon. They then coated the carbon nanotube transistor with a lipid bilayer—basically a double wall of oil molecules that cells use to separate their insides from the environment. The scientists didn't use an actual cell membrane, however. To this basic cellular structure, they added an ion pump—a biological device that pumps charged atoms of calcium, potassium and other elements into and out of the cell. Then they added a solution of adenosine tri-phosphate, or ATP, which fuels the ion pump.

The ion pump changes the electrical charge inside the cell, which then

changes the electrical charge going through the transistor, which the scientists could measure and monitor.

## Entangled LEDs

Scientists at Toshiba Research Europe and the Cavendish Laboratory of the University of Cambridge have developed a novel light source. The new device, called 'entangled light-emitting diode,' contains a nanometre-scale



Entangled light-emitting diode

quantum dot that produces entangled photons which can be used for quantum computing.

"The discovery is significant because it will allow electrical addressing of many entangled light emitters on a single chip, opening the path to ultra-powerful semiconductor processors based on quantum computation," said Andrew Shields, who directs the work at Toshiba Research Europe (Cambridge, England).

## Faster, smaller quantum computers

Researchers in China have successfully achieved teleportation up to 16 km using quantum entanglement of photons. This feat could pave the way for faster and smaller quantum-based computers and unbreakable, encrypted communication across the world. A teleported telephone call, based on this theory, would also be difficult to be tapped.

"This is the longest reported distance over which photonic teleportation has been achieved to date, more than 20 times longer than the previous implementation," said Cheng-Zhi Peng, one of the co-authors of the study and a scientist at University of Science and Technology of China and Tsinghua University in Beijing.

In science fiction, teleportation usually means transfer of matter from one point to another, more or less instantaneously—a quantum mechanical phenomenon. According to the theory, bits of light and matter can become entangled with one another and anything that happens to one particle will happen to the other, regardless of the distance or intervening matter. Chinese scientists managed to transmit change of state information from photon to photon over a distance of 16 km.

## Computer viruses can infect humans too

A chip implanted in the human body has the potential to infect and be infected by other systems and devices. This may have serious implications for implanted medical devices.

This is the finding of a research being conducted to draw attention to the lack of security in implantable devices, including medical ones. Mark Gasson, a senior research fellow at the University of Reading's School of Systems Engineering, found that a virus-infected chip implanted in his hand was able to contaminate external systems, as well as devices that connected to those systems.

The chip implanted in his left hand near the base of his thumb allowed him to gain access to a building, as well as to operate a designated mobile phone. The computer virus was then able to infect the building access system, as well as devices that connected to that particular system. In a separate experiment, the building access system was used to pass the virus onto the chip implanted in Gasson.

## World's first LED projector

In collaboration with 3LCD, Samsung claims to have launched the world's